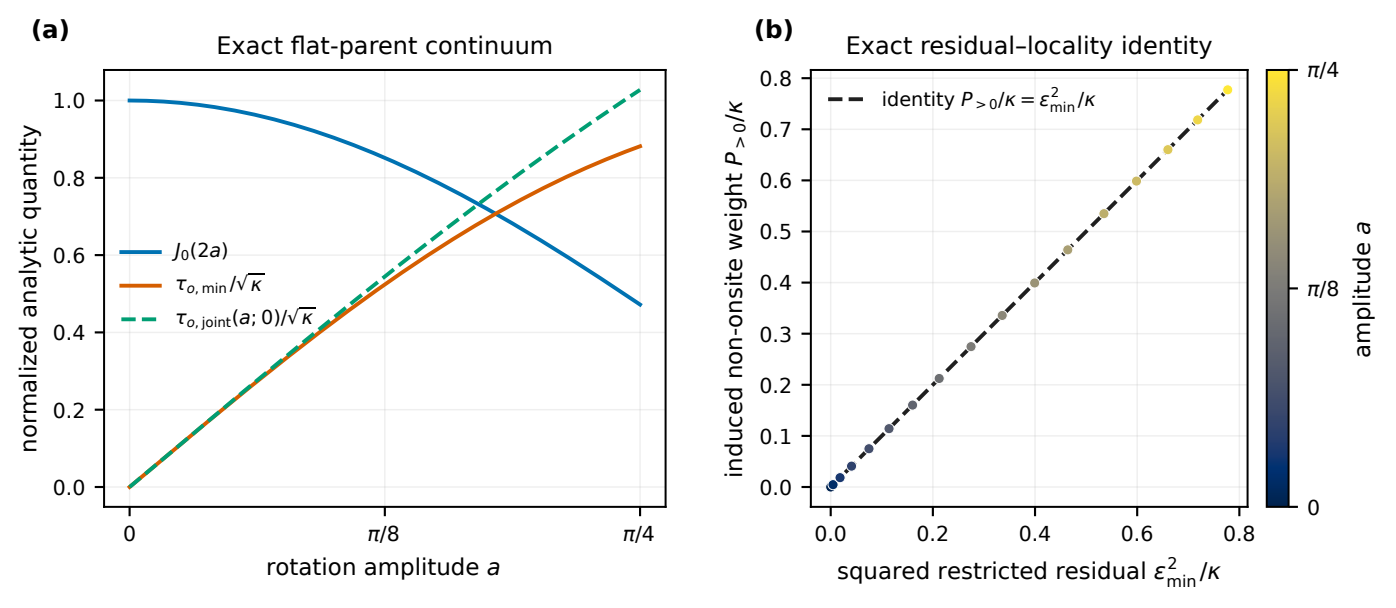
(a)
Exact flat-parent continuum
normalized analytic quantity
rotation amplitude a
$J_0(2a)$
$\tau_{o,\min}/\sqrt{\kappa}$
$\tau_{o,\mathrm{joint}}(a;0)/\sqrt{\kappa}$
0
π/8
π/4
(b)
Exact residual–locality identity
induced non-onsite weight $P_{>0}/\kappa$
squared restricted residual $\varepsilon^2_{\min}/\kappa$
identity $P_{>0}/\kappa = \varepsilon^2_{\min}/\kappa$
amplitude a